

PCR Amplification

Overview

A nucleic acid amplification technique to rapidly produce copies of DNA segments from a small initial sample.

★ Ex Taq DNA polymerase (Takara)

A. Materials

- a. 10x Taq Buffer
- b. 2.5mM dNTP
- c. 10 μ M Forward primer
- d. 10 μ M Reverse primer
- e. DNA Template (10 ng/ μ L)
- f. TaKaRa Ex Taq™
- g. PCR-grade water
- h. PCR machine

B. Procedure

1. Prepare reaction as follows

Reagent	1 Reaction (50 μ L)
10x Ex Taq Buffer	5 μ L
2.5 mM dNTP	4 μ L
10 μ M Forward primer	1 μ L
10 μ M Reverse primer	1 μ L
Template (10 ng/ μ L)	2 μ L
TaKaRa Ex Taq™	0.25 μ L
PCR-grade water	36.75 μ L

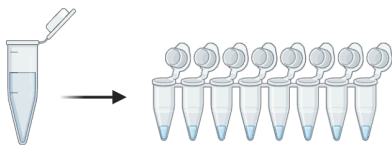
2. Incubate in the PCR machine and run the following thermal cycle.

Step		Temperature	Duration
Initial Denaturation		98°C	30 sec
25 Cycles	Denature	98°C	10 sec

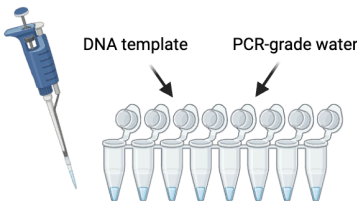
	Annealing	50–72°C	30 sec
	Extension	72°C	1 min/kb
Final Extension		72°C	2 min

PCR Amplification

- ① Prepare master mix and distribute to the each labelled PCR tubes.



- ② Add DNA template and PCR-grade water to each PCR tubes.



- ③ Incubate the PCR tubes into the PCR machine and run the thermal cycle.

